Rajveer Gupta

+91-8563938645 • rajveercareer1@gmail.com • LinkedIn • GitHub • Kaggle • Hugging Face

Portfolio: live demos, project case studies and the IEEE paper

Education

Indian Institute of Information Technology, Nagpur Aug 2024 – Jun 2028 (expected)
B.Tech, Computer Science & Engineering — CGPA: 9.08/10 *Nagpur, India*
Coursework: DSA, OOP (AA), Software Engineering (AA), Operating Systems (AB), Programming Language Design (AB), DBMS

Publication

A. Gupta, A. Singh, **R. Gupta**, P. Jain, T. Patidar, "Fault Tolerant Industrial Vision Framework Through Semantic-Guided Cross-View Temporal Reconstruction and Restoration," *IEEE IRAI 2026*, Melbourne (accepted and presented) | **Paper**

Experience

Research Intern, Computer Vision — IIIT Nagpur (Advisor: Dr. Nishat Afshan Ansari) 2026
Fault-tolerant industrial camera monitoring · Python, PyTorch, OpenCV
Overview: co-developed HocVid, which keeps a **3**-camera network working when one camera fails.

- Built camera-health detection as a **3**-state machine (healthy, degraded, dead) over a degradation score; a **16**-frame hysteresis window stops state flicker: **89.4%** accuracy.
- Routed each camera through one of **2** paths (direct restoration or full reconstruction) and added a quality gate that rolls back bad output frames to a safe fallback.
- Evaluated under injected physical faults (blur, lens dust, thermal noise, signal loss): **30.02 dB PSNR** on Wildtrack, +3.65 dB over the best naive cascade; **32.42 dB / 0.953 SSIM** on GoPro deblurring, **+2.37 dB** over MoCE-IR.

Projects

ReturnGuard — Multi-Agent Return-Fraud Detection for E-commerce 2026
GitHub | Live | Demo — Python, LangGraph, Gemini, PyTorch, DINOv2, Pydantic, Flask, SQLite, GitHub Actions
Overview: checks at the doorstep that a returned product is the unit that was delivered, before the refund.

- Designed a LangGraph state machine with parallel policy, history and vision agents and an LLM-free triage router: **79%** of returns need zero LLM calls (**0.21** per return, about **$1** per **1,000**).
- Added a blind label-reading agent, Pydantic-validated outputs, timeouts and retries with a fail-safe to human review; **35** offline tests run in CI, which auto-deploys to Hugging Face Spaces.
- Measured with confidence intervals: **100%** precision, **93.3%** recall and **0.986** F0.5 on **48** labeled pairs (v1: **90.9%** precision); **97.4%** precision at **100%** recall across **36** photo-distortion conditions.

Memora — Memory-Care App for Alzheimer's Patients 2025 – 2026
GitHub | Live | Results | Case study — Python, Flask, ONNX, PyTorch, RoBERTa · Sole developer
Overview: tells a patient who is in front of them in a loved one's voice and finds medicines; rebuilt solo from ARKA.

- Secured the app with AES-256-GCM encryption at rest, CSRF and XSS regression tests and a consent-gated AWS Rekognition fallback; **81** offline tests in CI and **53/53** end-to-end checks on the live deployment.
- Raised LFW face recognition from **87.1%** to **98.9%** (95% CI **98.2–99.5**) and cut latency from **395** to **21 ms** by fixing detection and alignment first (YuNet + SFace).

Stock Market Simulator — Object-Oriented Trading Engine in C++ 2025
GitHub — C++, STL, OOP, JSON, Python, Flask

- Designed a **7**-class engine (Trader, Company, Stock, Transaction, SIP, Market, Leaderboard) with encapsulation, inheritance, friend classes and a non-instantiable static `Market` utility.
- Decoupled the engine from the web layer: a Flask server drives the compiled C++ binary over stdin/stdout pipes, with **O(1)** holdings lookup in an `unordered_map` and JSON persistence.

Resolv — Multilingual Entity Resolution over 11.7M Records (Amazon ML Challenge 2026) 2026
*GitHub | Live | Demo — Team lead of 4 · **AIR 471 of 32K teams** · PyTorch, multilingual-e5, Qwen3, LightGBM*

- Built a multi-stage pipeline (blocking, dense retrieval, re-ranking, matching) over **11.7M** records, cutting **86.7** candidates per business to **5.58** and lifting leaderboard F0.5 to **0.986**.

Achievements

- **Led the team** in **Amazon ML Challenge 2026** to **AIR 471 of 32K teams**.
- **Led the team** in **Microsoft Imagine Cup 2025** to the **MVP Round** (**60K+** teams); managed the full ML pipeline.
- **Amazon ML Summer School 2026** (top **3,000** of **134,000+**) • Solved **250+** DSA problems (LeetCode, GfG).
- **Smart India Hackathon 2025**, **National Semi-Finalist** (AyurSutra) among **100,000+** participants.
- **Best Innovation Award**, IIT Hyderabad (Memora) • IEEE IES GenAI Hackathon Finalist.

Skills

Languages: C++, Java, Python, C, SQL
Core CS: DSA, OOP, System Design (REST APIs, microservices, parallel pipelines), OS, DBMS, Computer Networks
Backend & Tools: Flask, FastAPI, SQLite, PostgreSQL, Docker, Git, Linux, GitHub Actions, pytest
AI/ML: PyTorch, LangGraph, OpenCV, ONNX Runtime, scikit-learn, LightGBM